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B.E - (Computer) (Sem-VII)(Revised Course 2019-2020)
EXAMINATION DECEMBER 2023
Compiler Design

[Time: 3 Hours]**[Max. Marks:100]**

- Instructions:**
1. Answer any 5 questions by selecting 2 questions from Part-A and 2 questions from Part-B and 1 question from Part-C
 2. Make suitable assumption only if required

Part-A

- Q1 a. With the help of a neat diagram explain the phases of compiler. 12
- b. Give the DFA implementation of the lexical analyser for the following LEX program: 08
- ```
%{
enum {A,AB,ABB}
}%
%%
a {return (B);}
abb {return (ABB);}
a*b+ {return (ABB);}
%%
```
- Q2 a. What is left-recursion? How is left-recursion eliminated? Eliminate left-recursion from the following grammar:  $S \rightarrow Aa|b$  08  
 $A \rightarrow Ac|Sd$
- b. What is CFG? Explain with an example. 04
- c. Compute the FIRST and FOLLOW for the following context Free Grammar: 08
- ```
S → ABCDE
A → a|ε
B → B|ε
C → c
D → d|ε
E → d|ε
```
- Q3 a. Explain the applications of compiler technology. 05
- b. Write a lex program to validate identifiers and floating-point numbers. 05
- c. Write a short note on YACC 05
- d. Write a short note on top-down parsers 05

Part-B

- Q4 a. Construct operator precedence parsing table for the following grammar and parse the string: (a,(a,a)) **08**
- $S \rightarrow (L)|a$
 $L \rightarrow L, S|S$
- b. Construct SLR(1) parsing table for the following grammar: **08**
- $S \rightarrow aSbS$
 $S \rightarrow a$
- c. Translate the following expression into quadruples and triples: **04**
- $(x * y) * (p + q) + (x + y + z)$
- Q5 a. Explain the different issues in the design of code generator. **08**
- b. Explain the code generation algorithm with an example. **08**
- c. Write a short note on next use information. **04**
- Q6 a. Identify the basic blocks and draw the flow graph for the following source code. **08**
- total = 0.0
for (i=1; i<n; i++)
{
total=total +x[i];
}
b. What is peephole optimization? Discuss four peephole optimization techniques with the help of example. **08**
- c. What is Symbol table? Describe the contents of Symbol table. **04**

Part C

- Q7 a. Explain the move to higher level languages. **05**
- b. Explain the role of finite automata in lexical analysis **05**
- c. Consider the following grammar: **05**
- $E \rightarrow E + E | E - E | E * E | E / E | a | b$
Obtain leftmost and rightmost derivation for the string $a+b*a+b$
- d. Explain recursive descent parser **05**
- Q8 a. Explain the recovery techniques of lexical phase errors **05**
- b. Write translation scheme to produce three-address code for Boolean expressions **05**
- c. Explain the data structures used for symbol table **05**
- d. Explain the DAG representation of basic blocks **05**